Deliverable #1 Software Requirements Specification

SE 3A04: Software Design II – Large System Design

Tutorial Number: T01

Group Number: G06

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1 Introduction

This software requirements specifications (SRS) document will disclose the requirements for SCEMAS, an application that aims to provide smart city environmental monitoring as well that providing an accompanying alert system. It will discuss the purpose and scope of SCEMAS, user characteristics, business requirements and constraints, and will provide a use case diagram.

1.1 Purpose

The purpose of this SRS document is to provide a lens on the software requirements, use cases, and the characteristics of the user and environment of SCEMAS.

This SRS is intended to be viewed by stakeholders for SCEMAS, which includes archetypes such as project managers, developers, municipal level managers, and SCEMAS investors.

1.2 Scope

The Smart City Environmental Monitoring and Alert System (SCEMAS) is a software system that collects environmental sensor readings, including air quality, noise levels, temperature, and humidity, and converts them into information that people can use. It helps city operators see what is happening in near real time and it also shares high-level summaries with the public. SCEMAS achieves this by accepting sensor data through MQTT, checking that incoming data are in the correct format and within reasonable value ranges, and storing all valid telemetry in a database designed for time-series data [1]. After that, the system calculates zone-level summaries for areas in the city, like five-minute averages and hourly maximums, so users understand conditions without needing to interpret raw readings.

SCEMAS includes an operational dashboard for city operators, a public dashboard for residents, and a read-only public API that third parties can use to access approved aggregated data. The system also includes services for input data, storage, aggregation, and alerting. When environmental readings cross a predefined threshold, the alert engine creates an alert, stores it with a status and history, and makes it visible in the operational dashboard. Role-based access control (RBAC) is used to control which roles have which permissions, and the system keeps audit logs of important events like logins, failed login attempts, alert acknowledgments, and administrative changes. To keep public access safe, SCEMAS only exposes zone-level summary data rather than raw sensor readings and uses rate limiting to prevent over scraping. SCEMAS also protects data that is in transit using secure communication like TLS.

The main goals of SCEMAS are to improve awareness of environmental conditions in the city, support safety and sustainability by detecting unsafe conditions, and give both operators and the public summaries rather than hard-to-interpret raw data. A main objective is that alerts should show up for operators within 30 seconds from detection to visibility [1]. SCEMAS will not collect, store, or process personally identifiable information (PII), and the public API will be read only so external users cannot change system data [1]. The system focuses on monitoring, aggregation, and alerting based on thresholds rather than forecasting.

1.3 Definitions, Acronyms, and Abbreviations

- **API (Application Programming Interface):** An interface that allows software systems to communicate.
- **MQTT:** Message Queuing Telemetry Transport
- **PII (Personally Identifiable Information):** Information that can identify a specific person, like full name, phone number, and home address
- **RBAC (Role Based Access Control):** An access control method where permissions are assigned based on user roles, like General Public, City Operator, and Government Administrator.
- **REST:** A style for web APIs that returns structured responses often in JSON format
- **Telemetry:** Environmental readings using sensors, such as air quality, noise level, temperature, and humidity.
- **TLS (Transport Layer Security):** A protocol that encrypts data in transit to protect confidentiality.

1.4 References

[1] R. Khedri, "Project Outline for Software Design II (SE3A04) - Smart City Environmental Monitoring & Alert System (SCEMAS)," McMaster University, 09 January 2026. [Online]. Available: https://avenue.cllmcmaster.ca/d2l/le/lessons/727030/topics/5498994.

[2] E. Tuorinsky, "Guidance on NIST 800-171 log retention," DTS, 15 November 2022. [Online]. Available: https://consultdts.com/article/nist-800-171-log-retention/.

[3] W. Lidwell, J. Butler and K. Holden, The Universal Principles Of Design, Beverly, Massachusetts: Rockport, 2010.

[4] P. Tian, G. Xu, C. Han, X. Zheng, K. Zhang, C. Du, F. Wei and S. Zhang, "Effects of Paradigm Color and Screen Brightness on Visual Fatigue in Light Environment of Night Based on Eye Tracker and EEG Acquisition Equipment," *Sensors,* vol. 22, no. 11, p. 4082, 26 May 2022.

[5] D. S. McCrickard, M. Czerwinski and L. Bartram, "Introduction: design and evaluation of notification user interfaces," *International Journal of Human-Computer Studies,* vol. 58, no. 5, pp. 509 - 514, May 2003.

[6] M. Zuschlag, "How important is it to eliminate clicks?," User Experience Stack Exchange, 02 June 2013. [Online]. Available: https://ux.stackexchange.com/questions/40435/how-important-is-it-to-eliminate-clicks. [Accessed 12 February 2026].

[7] Environment and Climate Change Canada (ECCC), "Understanding Air Quality Health Index messages," Government of Canada, 14 November 2018. [Online]. Available: https://www.canada.ca/en/environment-climate-change/services/air-quality-health-index/understanding-messages.html .

[8] J. Nielsen, "Response Times: The 3 Important Limits," Nielsen Norman Group, 1 January 1993. [Online]. Available: https://www.nngroup.com/articles/response-times-3-important-limits/.

[9] S. Hyken, "The Personalized Customer Experience: Consumers Want You To Know Them," Forbes, 17 April 2024. [Online]. Available: https://www.forbes.com/sites/shephyken/2024/04/14/the-personalized-customer-experience-customers-want-you-to-know-them/.

[10] P. Ivanigova, "Why Timezones Are Confusing," Medium, 15 June 2025. [Online]. Available: https://medium.com/@petraivanigova/the-great-timezone-madness-why-half-the-world-is-always-wrong-about-what-time-it-is-8e6ede0469f8.

[11] K. Ramsundar, "6 reasons every website should have an FAQ page (+ examples)," Sparked Digital Inc, 18 November 2025. [Online]. Available: https://sparkeddigital.ca/6-reasons-why-every-website-should-have-an-faq-page-plus-examples-tips-for-creating-one/.

[12] M. Gergets, "Why is vocabulary important?," The Mind Company, 30 August 2025. [Online]. Available: https://themindcompany.com/blog/why-is-vocabulary-important.

[13] J. Nielsen, "10 Usability Heuristics for User Interface Design," Nielsen Norman Group, 30 January 2024. [Online]. Available: https://www.nngroup.com/articles/ten-usability-heuristics/.

[14] M. McCloskey, "Keyboard-Only Navigation for Improved Accessibility," Nielsen Norman Group, 06 April 2014. [Online]. Available: https://www.nngroup.com/articles/keyboard-accessibility/ .

[15] Government of Canada, "Designing for users with low vision," Government of Canada, 19 September 2025. [Online]. Available: https://a11y.canada.ca/en/designing-for-users-with-low-vision/ .

[16] Keyfactor, "Time Requirements," Keyfactor Inc., 2026. [Online]. Available: https://docs.keyfactor.com/signserver/latest/time-requirements. [Accessed 12 February 2026].

[17] GigaCloud, "SLA 99.95%: what this means and how the cloud operator is responsible," GigaCloud, 20 September 2022. [Online]. Available: https://gigacloud.eu/articles/sla-99-95-what-this-means-and-what-is-the-responsibility-of-the-cloud-operator/ .

[18] Amazon Web Services, "Failure management," Amazon, 2026. [Online]. Available: https://docs.aws.amazon.com/wellarchitected/latest/iot-lens/failure-management.html. [Accessed 12 February 2026].

[19] World Meteorological Organization, "Low-cost sensors can improve air quality monitoring and people's health," World Meteorological Organization, 13 June 2024. [Online]. Available: https://wmo.int/news/media-centre/low-cost-sensors-can-improve-air-quality-monitoring-and-peoples-health.

[20] S. Oloruntoba, A. S. Walia and M. Kurup, "SOLID Design Principles Explained: Building Better Software Architecture," DigitalOcean LLC., 11 June 2025. [Online]. Available: https://www.digitalocean.com/community/conceptual-articles/s-o-l-i-d-the-first-five-principles-of-object-oriented-design#open-closed-principle.

[21] C/S2ESC - Software & Systems Engineering Standards Committee, "ISO/IEC/IEEE International Standard - Systems and software engineering -- Life cycle processes -- Requirements engineering," IEEE Standards Association, 23 October 2018. [Online]. Available: https://standards.ieee.org/ieee/29148/6937/.

[22] ISO 25000, "ISO/IEC 25010," ISO 25000, 2026. [Online]. Available: https://iso25000.com/index.php/en/iso-25000-standards/iso-25010. [Accessed 12 February 2026].

[23] Atlassian, "What is IT support?," Atlassian, 2025. [Online]. Available: https://www.atlassian.com/itsm/itil. [Accessed 12 February 2026].

[24] K. Singh, "What is User Authentication, and Why is it Important?," LoginRadius Inc., 20 June 2025. [Online]. Available: https://www.loginradius.com/blog/identity/what-is-user-authentication.

[25] Qntrl Team, "What is role-based data access and why is it essential for your organization?," Zoho Corporation Pvt. Ltd., 15 November 2023. [Online]. Available: https://www.qntrl.com/blog/role-based-data-access.html.

[26] MicroAge, "Why Encrypt Your Data?," MicroAge, 2026. [Online]. Available: https://microage.ca/why-encrypt-your-data/ . [Accessed 12 February 2026].

[27] A. McCarvill, "An Introduction to the Importance of Input Validation in Preventing Security Vulnerabilities," Bright Security, 11 August 2023. [Online]. Available: https://brightsec.com/blog/an-introduction-to-the-importance-of-input-validation-in-preventing-security-vulnerabilities/.

[28] Google Play, "Google Play Developer Distribution Agreement," Google, 17 November 2020. [Online]. Available: https://play.google.com/about/developer-distribution-agreement.html..

[29] E. Mor-Samuels, "Everything you need to know about mobile app analytics," AppsFlyer Ltd., 29 October 2024. [Online]. Available: https://www.appsflyer.com/blog/measurement-analytics/everything-you-need-to-know-about-mobile-app-analytics/.

[30] Fortinet, Inc., "What Is SQL Injection?," Fortinet, Inc., 2025. [Online]. Available: https://www.fortinet.com/uk/resources/cyberglossary/sql-injection. [Accessed 12 February 2026].

[31] M. F. Ursu, V. Zsombori, J. Wyver, L. Conrad, I. Kegel and D. Williams, "Interactive documentaries: A Golden Age," *Computers in Entertainment (CIE),* vol. 7, no. 3, pp. 1-29, September 2009.

[32] ROTEL Project, "Designing User Experiences For All," ROTEL Project, December 2024. [Online]. Available: https://rotel.pressbooks.pub/culturally-responsive-computing/chapter/designing-user-experiences-for-all/.

[33] Computer Security Resource Center, "Role Based Access Control RBAC," National Institute of Standards and Technology, 08 May 2025. [Online]. Available: https://csrc.nist.gov/projects/role-based-access-control.

[34] Government of Canada, "What and how to collect personal information," Government of Canada, 08 November 2023. [Online]. Available: https://www.canada.ca/en/government/system/digital-government/digital-privacy-playbook/what-how-to-collect.html. [Accessed 12 February 2026].

[35] Merlin, "Reasons Why Logging is Crucial," Erstream, 13 July 2022. [Online]. Available: https://www.merlincdn.com/blog/5-reasons-why-logging-is-crucial-1.

[36] Okta, "The Advantages and Benefits of Single Sign-On (SSO)," Okta, 28 April 2022. [Online]. Available: https://www.okta.com/en-gb/blog/identity-security/benefits-of-single-sign-on/ .

1.5 Overview

Section 2 contains the overall project description, which discusses the product perspective, functions, user characteristics, constraints, assumptions and dependencies, and requirements that can be deferred to later releases. Section 3 contains a use case diagram for the system, highlighting the ideal flow of the application. Section 4 highlights the functional requirements of various business events and viewpoints. Section 5 contains the non-functional requirements for the system, where requirements such as look and feel, usability and humanity, performance, operational and environmental, maintainability and support, security, cultural and political, and legal requirements are present. Section 6 states the innovative features that our project consists of that make it unique. Finally, Section A indicates the division of labour between the team members for the development of this SRS.

2 Overall Product Description

2.1 Product Perspective

SCEMAS is a software platform that aims to collect, process, and invite action upon environmental data from a distributed sensor network. Its main purpose is to turn raw measurement into high-level data that can be actionable by both municipal and civilian level users. SCEMAS is not unlike other geographically based applications, namely software like *Fulcrum* or even to government-operated ones like *WeatherCAN.* The product will provide users with an RBAC dashboard that will display information translated from raw environmental data, and will provide sensor locations, system health, and historical data amongst other information. The style of information presented to the user is comparable to *Microsoft Power BI* and *Google Looker Studio*. SCEMAS will include a public, read-only REST API that provides third parties with various non-sensitive data. This is reminiscent of a common practice in many commercially available software, like *GitHub, Spotify,* and *YouTube.*

SCEMAS is planned to interact with the larger system of physical sensors and is expected to be flexible in the types it can work with. The product will aim to maintain functionality with commercially available environment sensors, fitting within its constraints to provide the expected functionality.

SCEMAS will require an interface that allows it to interact and transfer information with the sensor hardware, as well as an interface that allows SCEMAS to accurately detect geological positions relative to the user.

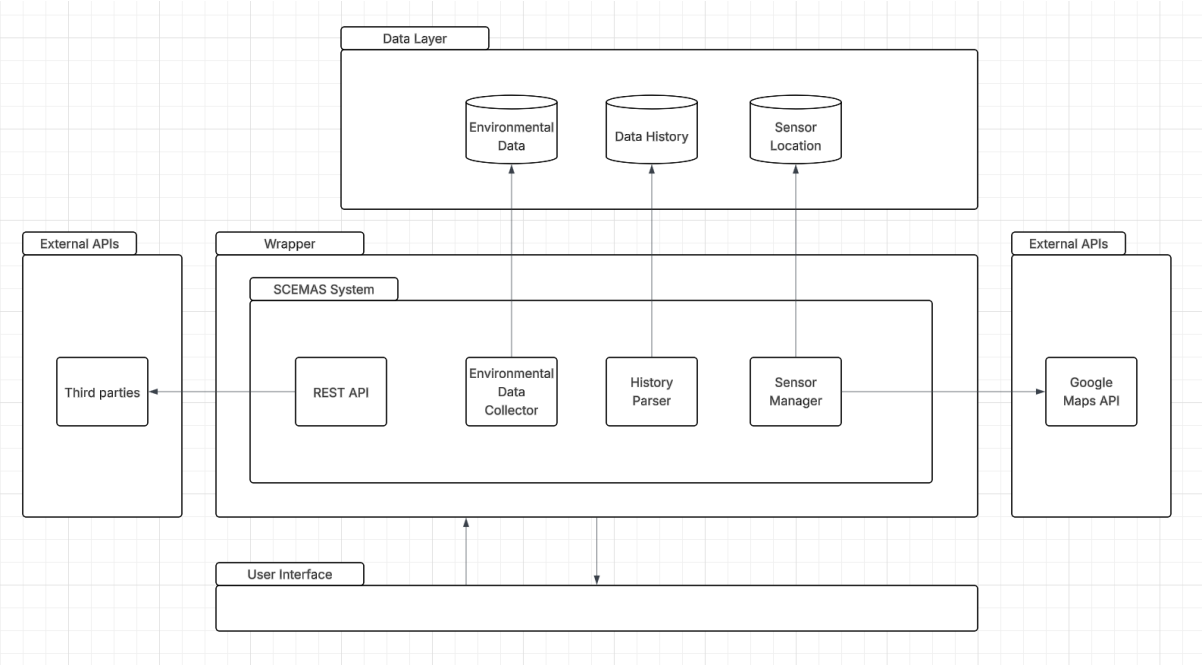

Figure 1: System Diagram of SCEMAS

2.2 Product Functions

SCEMAS focuses on continuously collecting environmental readings around a city, turn raw telemetry into zone-level summaries, and raise alerts for unsafe or unusual conditions. *Table 1* summarizes the main systems parts and what each one does at a high level.

Table 1: High level description of the major system components

Major System Part	**Functions**
Telemetry Service	• Accept telemetry streams using MQTT • Protect data in transit using TLS • Validate telemetry for correct formats and reasonable value ranges, while rejecting invalid data • Store all valid readings in a time-series database for real-time and historical use
Aggregation Service	• Continuously compute summaries like averages and maximums for city zones • Provide aggregated results to the dashboards, the public API, and alert evaluation
Alerting Service	• Let administrators define alert conditions based on unusual readings • Check telemetry and zone summaries in real time to check for violation and trigger alerts • Record alerts with status and history for tracking and review
Dashboard System	• Provide a secure operational dashboard with maps, alerts, and historical trends • Support alert acknowledgment and response workflows. This includes visibility of critical alerts within ~30 seconds of detection • Offer a public dashboard that shows high-level summaries for residents
Public API Service	• Provide a read-only REST API that shares approved aggregated zone-level data • Limit how often users can call the API to prevent overuse (rate limiting)

	• Provide documentation for the API such that a developer can understand
Security System	• Use RBAC so operators and admins have different permissions • Require authentication for non-public features • Keep logs of actions like logins, failed logins, role changes, and alert acknowledgements • Support privacy goals by avoiding the collecting of PII
Messaging Service	• Allows Data Analysts and Government Administration to send and receive messages to each other inside SCEMAS • Stores messages with timestamp, sender role, and receiver role. • Track message status such as sent, delivered, and read.

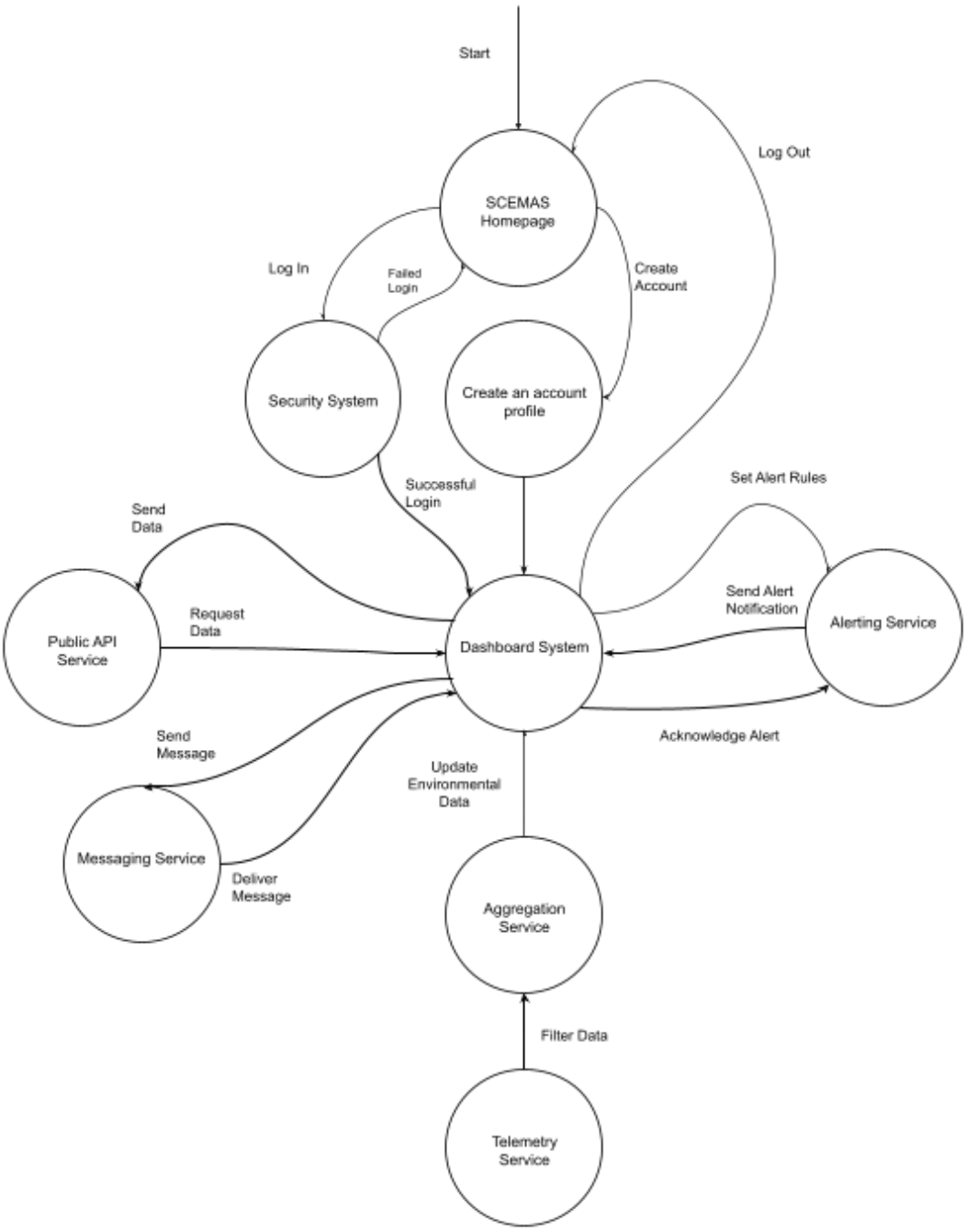

Figure 2: State Diagram of the SCEMAS

2.3 User Characteristics

Table 2: General Characteristics of various users of SCEMAS

Users	**Characteristics**
Data Analysts (SCEMAS)	Users that are utilizing the app for data purposes would be expected to hold an education in data science/analytics with some prior experience. Moderate technical expertise is required.
General Public	For these users the only requirement is basic literacy skills. No experience necessary and a basic knowledge of operating a computer is required.
Government Staff	University/college education in urban planning, environmental science, public safety, or a related field. These users must complete a training module provided by SCEMAS prior to being granted access to the system. Must have some experience working with municipal tools. Moderate technical expertise is required.
City Operators	No formal education is required; however, these users must complete a training module provided by SCEMAS prior to being granted access to the system. Additionally, users are expected to possess basic knowledge of operating a computer.
External Developer	Expected to have a university/college education in tech, with some prior experience working with APIs. These users must complete a training module provided by SCEMAS prior to being granted access to the system

2.4 Constraints

1. Budget: The amount of money that can be spent building SCEMAS will limit what resources and technologies can be put into the project.
2. Project time: The time allocated to working on the project will affect the scale to which it will be built. It will also affect the resources that are available for the project.
3. Regulatory constraint: SCEMAS must comply with Canadian data protection and public sector data management policies. This may restrict how environmental data is stored and transmitted.
4. Sensors constraint: SCEMAS must support MQTT-compatible sensors and infrastructure that can communicate using the MQTT protocol

2.5 Assumptions and Dependencies

- Assumptions
    - SCEMAS will operate within a single city environment rather than supporting multiple cities simultaneously.
    - The deployment context is assumed to be within Ontario, Canada, and the system is not currently designed for international legal or regulatory variability.
    - SCEMAS will only support environmental metrics related to air quality, temperature, humidity, and noise levels in the initial release.

    - External developers will not require SCEMAS accounts and will instead access the system exclusively through the public REST API.
    - Sensor companies will not have full dashboard access and will instead interact with SCEMAS through external communication methods.
    - Historical telemetry and alert data will be retained for a minimum of 90 days rather than indefinitely [2].
    - The system will prioritize threshold-based alerting rather than advanced predictive analytics or machine learning forecasting in the initial release.
    - The architecture will assume a cloud-native deployment model rather than on-premise hosting.
    - The system will aggregate and display environmental data at zone level only and will not expose raw sensor coordinates to public users.
    - Government administrators are solely responsible for defining and activating environmental alert thresholds. City operators may acknowledge alerts but cannot modify thresholds.
- Other assumptions that, if it fails to hold, could require a change to the requirement
    - Assume the APIs are fully functional during the operation of the product.
    - Assume the APIs are safe and secure.
    - Third-party services used for visualization and authentication are available and reliable.
    - Cloud infrastructure is available for hosting the system.

2.6 Apportioning of Requirements

1. External system integrations such as with emergency services and auto-dispatches.
    a. This is a feature that would streamline the process of obtaining the alert and taking appropriate action in a quick manner. In the initial release, the user will have to use a separate system to provide law enforcement commands and will only have the ability to acknowledge alerts and log appropriate actions.
2. Support for additional environmental events such as water quality, radiation, and earthquake
    a. The initial release will only monitor environmental indicators such as air quality, noise levels, temperature, and humidity.

3 Use Case Diagram

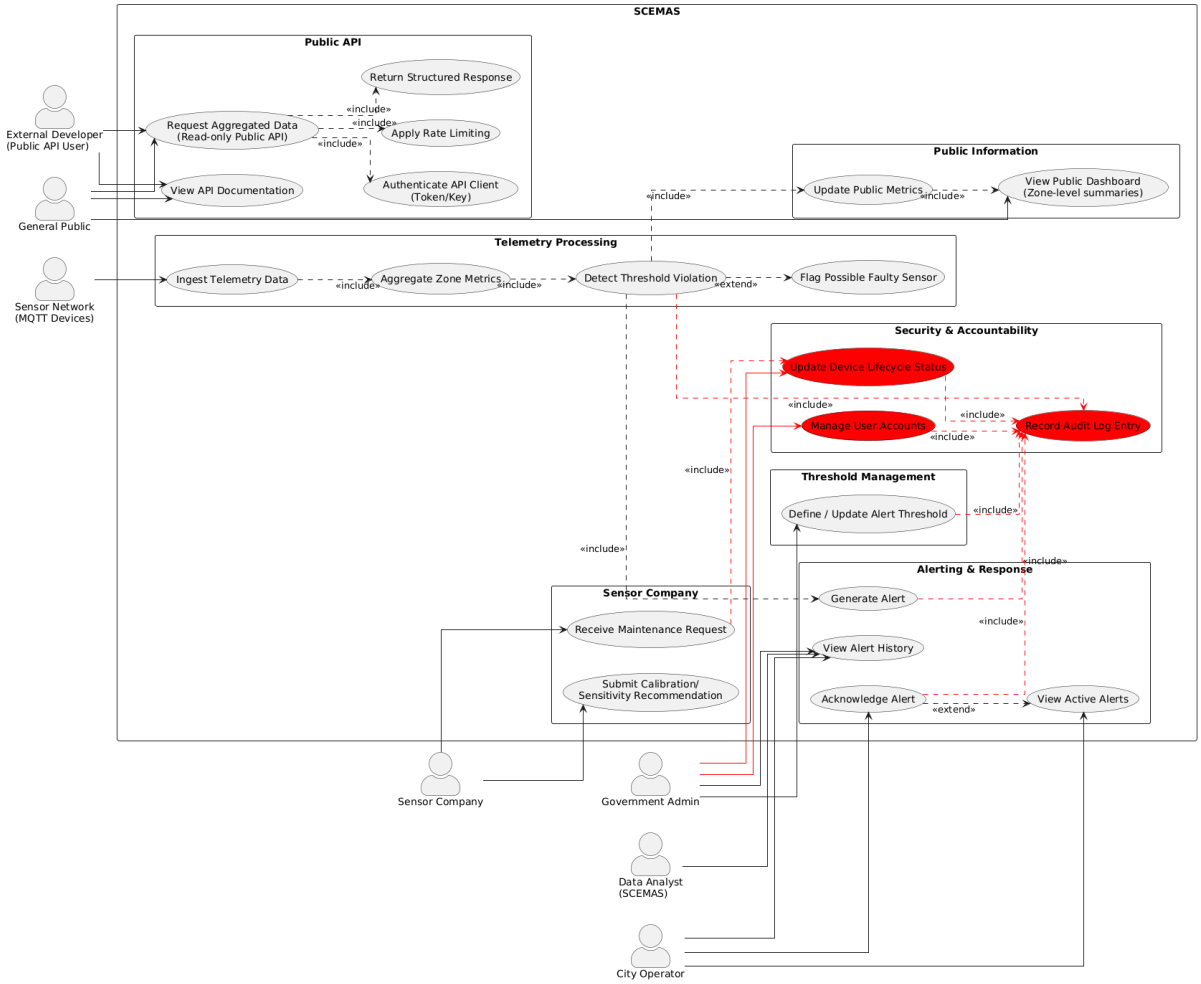

Figure 3: Use Case Diagram for the SCEMAS

4 Highlights of Functional Requirements

Business events to be considered will include:

- BE1: Login to dashboard
- BE2: Define alert thresholds
- BE3: Threshold violations
- BE4: Create Account
- BE5: Edit Account within limit
- BE6: Handling Public API requests
- BE7: Anomalous Trend Violations
- BE8: Validating data

Viewpoints to be considered will include:

- VP1: General Public
- VP2: City Operators
- VP3: Government (Admin)
- VP4: Sensor Company
- VP5: Data Analysts (SCEMAS)
- VP6: External Developers

BE1. Login to Dashboard

Pre-Condition: The user already has a SCEMAS account with an assigned role and the login service is available.

VP1. General Public

Main Success Scenario:

1. User opens the SCHEMAS public dashboard
2. User selects Log In
3. System displays the login screen
4. User enters username/email and password
5. System verifies the credentials
6. System identifies the user role as General Public
7. System redirects the user to the public dashboard view with additional logged in features
8. The system records the login event in the audit logs

Secondary Scenario:

5i. Invalid credentials
  5i.1. System shows a failed login/error message and does not log the user in
  5i.2. System allows user to retry login
5ii. Login service is down/unavailable
  5ii.1. System shows an error message and instructs the user to try again later

VP2. City Operators

Main Success Scenario:

1. Operator opens SCEMAS and selects Log In

2. System displays Log In screen
3. Operator enters username/email and password
4. System verifies the credentials
5. System classes user as City Operator role
6. System requires additional authentication due to extra permissions
7. Operator completes extra authentication
8. System allows access and redirects operator to the operational dashboard
9. The system records the login and verification result in the audit logs

Secondary Scenario:

4i. Invalid Credentials
    4i.1. System shows an error message and allows retry
6i. Additional authentication fails
    6i.1. System denies access
    6i.2. System makes logs about failed attempt
    6i.3. System redirects operator to Login screen
8i. Operational dashboard unavailable
    8i.1. System logs the successful login
    8i.2. System displays a message that the operational dashboard is currently unavailable

VP3: Government (Admin)

Main Success Scenario:

1. Administrator opens SCEMAS and selects Log In
2. System displays the login screen
3. Administrator enters username/email and password
4. System verifies the credentials
5. System classifies user as Government admin
6. System requires extra authentication to allow access
7. Administrator completes the verification process
8. System allows access and redirects the administrator to the admin view of the operational dashboard
9. System records the login and verification result in the audit logs

Secondary Scenario:

4i. Invalid login
    4i.1. System shows error message and allows retry
6i. Authentication Fails
    6i.1. System denies access
    6i.2. System makes log of the event as suspicious activity
    6i.3 System returns the administrator to the login page

VP4: Sensor Company

N/A - Sensor company is outsourced and do not have SCEMAS accounts

VP5: Data Analyst (SCEMAS)

Main Success Scenario:

1. Analyst opens SCEMAS and selects Log In

2. System redirects to the login screen
3. Analyst enters username/email and password
4. System verifies credentials
5. System identifies the user role as Data Analyst
6. System allows access and redirects user to analytics and trend's view
7. The system records the login event in the audit logs

Secondary Scenario:

4i. Invalid credentials
    4i.1 System shows error message and allows retry
6i. Trends page not available
    6i.1. System logs the login
    6i.2. System displays message that page is currently not available

VP6: Developer (SCEMAS)

N/A - External developers access the public API without logging into SCEMAS

Global Scenario:

Pre-condition: The user has a valid account and SCEMAS login service is operational

Main Success Scenario:

1. User selects Log In on SCEMAS
2. System displays the login screen
3. User enters login credentials
4. System verifies the credentials
5. System classes user's role
6. If the role has additional permissions, system requests for additional authentication
7. System grants access based on RBAC permissions
8. System redirects the user to the correct page/dashboard based on their role
9. System records the login event in the audit logs

Secondary Scenario:

4i. Credentials are invalid
    4i.1. System denies access and allows user to retry
6i. Additional verification fails
    6i.1. System denies access and logs the event as suspicious
8i. Target dashboard view is unavailable
    8i.1. System logs the successful login
    8i.2. System displays message saying that the page is currently unavailable

BE2. Defining alert thresholds

Pre-condition: An authorized user is already logged into SCEMAS, and at least one environmental alert threshold has been defined and enabled.

VP1. General Public

N/A

VP2. City Operators

N/A

VP3. Government (Admin)

1. Government Administrator determines that a new environmental alert threshold is required due to external policy, regulatory guidance, or public safety considerations.
2. System provides access to the SCEMAS administrative dashboard.
3. Government Administrator accesses the administrative dashboard.
4. System verifies the administrator's role and permissions.
5. Government Administrator navigates to the Alert Management section.
6. System displays existing alert thresholds and an option to create a new threshold.
7. Government Administrator selects Create Alert Threshold.
8. System displays the alert threshold configuration form.
9. Government Administrator selects the environmental metric (e.g., air quality, temperature, noise).
10. System displays configuration options for the selected metric.
11. Government Administrator selects the geographic zone to which the threshold applies.
12. System confirms the selected geographic zone.
13. Government Administrator defines the threshold condition and severity level.
14. System validates the entered information.
15. Government Administrator submits the alert threshold configuration.
16. System stores the alert threshold and marks it as active.
17. Government Administrator reviews the confirmation.
18. System records the action in the audit logs and confirms successful creation of the alert threshold.

Secondary Scenario:

9i. Unsupported environmental metric selected
    9i.1 System displays an error message.
    9i.2 Government Administrator corrects the metric selection.
    9i.3 System returns to the configuration form.
13i. Invalid threshold value entered
    13i.1 System displays a validation error.
    13i.2 Government Administrator corrects the threshold value.
    13i.3 System re-validates the updated information.
16i. System fails to store the alert threshold
    16i.1 System logs the failure.
    16i.2 Government Administrator receives notification that creation failed.
    16i.3 System prompts the administrator to retry or cancel.

VP4: Sensor Company

Main Success Scenario:

1. Sensor Company Representative detects a change in sensor firmware or hardware that may affect environmental readings.
2. System provides a mechanism for external stakeholders to submit technical recommendations.
3. Sensor Company Representative prepares a technical recommendation describing the sensor behaviour.

4. System accepts recommendation submissions for SCEMAS review.
5. Sensor Company Representative submits the recommendation through the designated submission channel.
6. System receives the recommendation and records it for SCEMAS Data Analysts.

Secondary Scenario:

N/A

VP5. Data Analysts (SCEMAS)

Main Success Scenario:

1. System notifies the SCEMAS Data Analyst that a new alert threshold has been activated.
2. SCEMAS Data Analyst accesses the alert threshold configuration.
3. System displays the updated threshold parameters.
4. SCEMAS Data Analyst updates internal aggregation and anomaly-detection algorithms to align with the new threshold logic.
5. System validates the configuration changes.
6. SCEMAS Data Analyst deploys the updated algorithm configuration.
7. System applies the updated analytics logic to incoming telemetry streams.
8. SCEMAS Data Analyst reviews the deployment status.
9. System logs the configuration update in the audit log.

Secondary Scenario:

5i. Algorithm configuration invalid
    5i.1 System displays validation error.
    5i.2 SCEMAS Data Analyst corrects the algorithm configuration.
    5i.3 System re-validates the updated configuration.
7i. Deployment failure
    7i.1 System logs the failure.
    7i.2 SCEMAS Data Analyst is notified of the deployment issue.
    7i.3 System restores the previous algorithm configuration.

VP6: External Developers

N/A

Global Scenario:

Pre-condition: An authorized user is already logged into SCEMAS, and at least one environmental alert threshold has been defined and enabled.

Main Success Scenario:

1. Government Administrator defines and activates an environmental alert threshold.
2. System stores the alert threshold and enables monitoring for the defined condition.
3. Environmental Sensors transmit telemetry data to the system.
4. System receives and processes the incoming telemetry data.
5. Environmental Sensors continue transmitting updated environmental readings.
6. System evaluates the telemetry data against the defined alert thresholds.

7. Authorized User accesses the system to monitor environmental conditions.
8. System detects a threshold violation and generates an alert.
9. Authorized User views the generated alert.
10. System displays the alert details and records all actions in audit logs.

Secondary Scenario:

4i. Telemetry data is invalid or unavailable

4i.1 Environmental Sensors transmit corrupted or incomplete telemetry data.
4i.2 System detects that the telemetry data is invalid.
4i.3 System Administrator reviews the telemetry data issue.
4i.4 System logs the telemetry failure and continues monitoring incoming data.

8i. System fails to generate an alert

8i.1 Authorized User expects an alert notification based on environmental conditions.
8i.2 System fails to generate the alert due to an internal processing issue.
8i.3 System Administrator investigates the alert generation failure.
8i.4 System logs the failure and attempts to recover the alert evaluation process.

BE3. Threshold Violations

Pre-condition: At least one alert threshold has been defined and enabled, the telemetry data is being actively received and processed by the system, and the alerting system is operational.

VP1: General Public

Main Success Scenario:

1. System detects that a defined environmental threshold has been violated.
2. System updates public metrics for the affected areas.
3. Public dashboards reflect the updated metrics and use high level indicators for key public safety events.
4. Public REST API exposes aggregated data for the affected area.

Secondary Scenario:

N/A

VP2: City Operators

Main Success Scenario:

1. System detects that a defined environmental threshold has been violated.
2. System generates alert with associated metric, severity, areas, and timestamp.
3. System records alert and marks it active
4. Dashboard shows updated data in 30 seconds
5. City Operator access the dashboard
6. Operator selects the alert
7. System displays alert information
8. Operator acknowledges the alert
9. System updates alert status to acknowledged and updates audit logs.

Secondary Scenario:

4i. Dashboard is unavailable
    4i.1. Alert reminds active and is logged
    4i.2. Threshold violation handling continues when the dashboard is restored.
8i. Operator does not acknowledge alert and the alert remains active
9i. System fails to update alert status
    9i.1. Alert status remains unchanged
    9i.2 System logs failure for investigation.

VP3: Government (Admin)

Main Success Scenario:

1. System detects that a defined environmental threshold has been violated.
2. System logs threshold violation and assigns it to a particular role.
3. Administrator accesses alert history and status through administrator views.

Secondary Scenario:

N/A

VP4: Sensor Company

Main Success Scenario:

1. System detects that a threshold violation due to potentially faulty sensor readings.
2. System logs sensor meta data and alerts users with appropriate permissions.

Secondary Scenario:

N/A

VP5: Data Analysts (SCEMAS)

Main Success Scenario:

1. System detects that a defined environmental threshold has been violated.
2. System records violation in alert history database
3. Analyst accesses past data for analysis and reporting
4. System provides up-to-date and historical context for the violation.

Secondary Scenario:

N/A

VP6: External Developers

Main Success Scenario:

1. System detects a defined environmental threshold has been violated.
2. System updates aggregated zone-level metrics.
3. Public REST API makes the updated metrics available.
4. External developer's application sends valid API request for zone data.
5. System authenticates and rate-limits the request according to public API policies.
6. System returns structured, read-only environmental data including updated status indicators (e.g. severity level, zone status, timestamps).
7. External developer's application processes and uses the updated environmental data.

Secondary Scenario:

4i. API request exceeds rate limit
    4i.1. System returns rate limit error message
    4i.2. System logs the rate limit error event.
4ii. API service temporarily unavailable.
    4ii.1. System returns service unavailable error message
    4ii.2. System logs the API outage event.
5i. Invalid API request
    5i.1. System rejects request
    5i.2. System returns structured error response.

Global Scenario:

Pre-condition: At least one alert threshold has been defined and enabled, the telemetry data is being actively received and processed by the system, and the alerting system is operational.

Main Success Scenario:

1. System receives and validates telemetry data.
2. System computes area specific metrics.

3. System evaluates metrics against defined environmental thresholds.
4. System detects that a defined environmental threshold has been violated.
5. System generates and logs an alert
6. Operational dashboard displays alert within 30 seconds.
7. System updates public metrics for the affected areas
8. City Operator acknowledges the alert
9. System updates alert status and records action

Secondary Scenario:

6i. Dashboard is unavailable
    6i.1. Alert remains active until dashboard access is restored
8i. Operator does not acknowledge the alert
    8i.1. Alert remains active until acknowledged.
9i. System fails to record acknowledgement.
    9i.1. Alert system remains unchanged and failure is logged.

BE4. Create Account

Pre-condition: Account creation is enabled by SCEMAS, and the account service is available.

VP1. General Public

Main Success Scenario:

1. User opens the SCEMAS public dashboard
2. User selects Create Account
3. System displays the account creation page
4. User enters information for public account like email and password
5. System validates inputs
6. System creates the account with General Public role
7. System confirms account creation
8. System records the account creation event in the audit logs
9. System redirects user to Log In screen

Secondary Scenario:

5i. Missing or invalid information
    5i.1. System shows validation errors and asks the user to correct/add them
6i. Account already exists for the email provided
    6i.1. System tells user that account exists and asks if they want to log in
6ii. System fails to create account
    6ii.1. System shows an error message and does not create the account

VP2: City Operators

Main Success Scenario:

1. Operator selects Create Account
2. System displays the account creation page for City Operator
3. Operator enters required information
4. System validates inputs
5. System redirects request for approval by administrator
6. Administrator approves the creation request

7. System creates the account with City Operator role
8. System informs user that account is approved
9. The system records the request and approval in the audit logs
10. Operator proceeds to log in page

Secondary Scenario:

4i. Missing or invalid information
    4i.1. System shows validation errors and asks for fixes
6i. Administrator rejects the request
    6i.1. System marks the request as rejected
    6i.2. System notifies operator of rejection
    6i.3. System records the decision in audit logs
6ii. Approval service not available
    6ii.1. System keeps the request but tells the operator of delays
    6ii.2. System records the pending request

VP3: Government (Admin)

Main Success Scenario:

1. Government user selects Create Account
2. System displays an admin account request page
3. Government user enters required information
4. System validates the inputs
5. System sends the request for verification due to high clearance
6. Verification is successful
7. System creates the account with Government Admin role
8. System confirms account creation
9. The system records the request and verification result in the audit logs
10. The system redirects the user to the login page

Secondary Scenario:

4i. Missing or invalid information
    4i.1. System shows validation errors and asks for fixes
6i. Verification fails
    6i.1. System denies the request and logs the request
    6i.2. System records the failed attempt in the audit logs
6ii. Verification not available
    6ii.1. System notifies the user that the request is delayed
    6ii.2. The system records the request as pending

VP4: Sensor Company

N/A - Sensor Company is outsourced and do not have SCEMAS accounts

VP5: Data Analyst (SCEMAS)

Main Success Scenario:

1. Analyst selects Create Account on SCEMAS
2. System displays an account creation page

3. Analyst enters required information for data analyst role
4. System validates the inputs
5. System sends the request for approval by an administrator
6. Administrator approves the analyst request
7. System creates the account with Data Analyst role
8. System notifies the analyst that the account is ready
9. System records the request and verification result in the audit logs
10. System redirects the analyst to the login page

Secondary Scenario:

4i. Missing or invalid information
    4i.1. System shows errors and asks for fixes
6i. Administrator rejects the request
    6i.1. System marks the request as rejected
    6i.2. System notifies the analyst that the request was rejected
    6i.3. System records the decisions in the audit logs

VP6: External Developers

N/A - External developers use the public API without creating a SCEMAS account

Global Scenario:

Pre-condition: Account creation is enabled and SCEMAS account services are available.

Main Success Scenario:

1. User selects Create Account on SCEMAS
2. System displays the appropriate account creation page based on requested role
3. User enters required information
4. System validates the inputs
5. If the role is high importance, the system sends the request for approval or verification
6. Approval or verification succeeds
7. System creates the account with the correct role and RBAC permissions
8. System confirms account creation and prompts the user to Log In
9. System records the account creation event in audit logs.

Secondary Scenario:

4i. Information is missing or invalid
    4i.1. System shows the errors and prompts correction
6i. Authentication is rejected/failed
    6i.1. System does not create the account and tells the user
    6i.2. System records the failure in the audit logs
7i. System fails to create the account
    7i.1. System shows an error and logs the failure

BE5. Editing account within limit

Precondition: The user has an unedited account, meaning it is in a default state. The user is already registered for this software.

VP1: General Public

Main Success Scenario:

1. User accesses SCEMAS software and enters login details.
2. System authenticates user.
3. User navigates to account management page.
4. System provides edits to the account that are available to this user's clearance level.
5. Once the user is done, they save their desired changes.
6. User account changes are properly reflected in the software database.

Secondary Scenario:

2i. System fails to authenticate user, returns to login page
    2i.1. User is given choice to try again
4i. System provides edit access to information available to higher clearances only
    4i.1. User is denied edit access, returns to previous page
6i. Changes are not properly reflected in database
    6i.1. System rolls data back to previous safe point

VP2: City Operators

Main Success Scenario:

1. Operator enters login details
2. System tries to authenticate user, but notices a higher clearance than public so it prompts for extra verification
3. Operator is authenticated, navigates to account management page
4. System provides edits to account that are available to this clearance level
5. Once done, the operator saves their desired changes
6. Account changes are properly reflected in the software database

Secondary Scenario:

2i. System fails extra verification, returns operator to login and flags interaction as suspicious
    2i.1. User is denied access, returned to login page
6i. Changes not properly reflected in database
    6i.1. System rolls back to previous stable state

VP3: Government (Admin)

Main Success Scenario:

1. Government associate accesses SCEMAS software, enters login details
2. System tries to authenticate user, but notices the utmost clearance level, and interacts with a private API with the government to confirm identity
3. User is authenticated, navigates to account management page
4. System provides mostly every edit possible to the account

5. Once done, the user saves their desired changes
6. Account changes are properly reflected in the software database

Secondary Scenario:

2i. System fails extra verification
    2i.1. Returns user to login and flags interaction as highly suspicious and creates a report in the government database
6i. Changes not properly reflected in database
    6i.1. Rolls back to previous stable state

VP4: Sensor Company

N/A

VP5: Data Analysts (SCEMAS)

N/A

VP6: External Developers

N/A

Global Scenario:

Precondition: The user has an unedited account, meaning it is in a default state. The user is already registered for this software.

Main Success Scenario:

1. User accesses SCEMAS software, enters login details.
2. System authenticates user in some fashion.
3. User navigates to account management page.
4. Once the user is done making changes, they save their edits.
5. User account changes are properly reflected in the software database.

Secondary Scenario:

2i. System fails to authenticate user
    2i.1. Return user to login page
4i. System provides edit access to information available to higher clearances only
    4i.1. User is allocated appropriate actions
5i. Changes are not properly reflected in database
    5i.1. System rolls back to previous safe state

BE6. Handling public API requests

Precondition: An external software gives an API request, with the interface already complete.

VP1: General Public

N/A

VP2: City Operators

N/A

VP3: Government

Main Success Scenario:

1. Government representative determines which information is exposed through the public API.
2. After confirming the correct information, approve the API exposure through system.

Secondary Scenario:

1i. Incorrect level of information is exposed.
    1i.1 The API is denied from functionality and is returned to developer to redo.

VP4: Sensor Company

N/A

VP5: Data Analysts (SCEMAS)

Main Success Scenario:

1. Data analyst finds information that could be provided by the API.
2. Analyzes the use case of the API, and how it relates to their work.
3. System confirms data integrity.
4. User moves onto utilizing the information provided by the API.

Secondary Scenario:

2i. API does not relate to their work.
    2i.1 Data analyst finds a different public API to assist them
3i. Data cannot have its integrity confirmed.
    3i.1 API is left unused.

VP6: External Developers

Main Success Scenario:

1. The developer researches the API and determines if it is useful.
2. They make an API request in the code.
3. API request is approved by system, providing relevant services to the developer software.
4. The external software will use the information, and create a report on how it was used, and which parts were utilized.

Secondary Scenario:

2i. API request is lost.
    2i.1. Sends back an error message with logs
3i. API request denied.
    3i.1. Return an error message to external entity
4i. The external software does not utilize the API.
    4i.1. Return an empty report

Global Scenario:

Precondition: An external software gives an API request, with the interface already complete.

Main Success Scenario:

1. Some sort of information is desired from SCEMAS, create an API request.
2. SCEMAS provides the API methods and software, allowing external applications to utilize it.
3. The API provides documentation, and constraints placed on usage of the API.
4. Once finished, sends a report back to SCEMAS backend based on usage stats.

Secondary Scenario:

1i. API request is lost/some malfunction happens.
    1i.1 Error report is produced and returned
3i. API is not utilized.
    3i.1. An empty error report is returned.

BE7. Anomalous trend violations

Pre-Condition: An anomalous threshold has been defined and enabled.

VP1: General Public

Main Success Scenario:

1. User opens SCEMAS and logs in as member of the general public.
2. System authenticates user.
3. User accessed the operational dashboard.
4. Dashboard displays a message stating that an anomalous trend violation is in effect.
5. User clicks on the message and views details.
6. System displays violation details.

Secondary Scenario:

2i. System fails to authenticate user.
4i. No anomalous trend violations are in effect.

VP2: City Operators

Main Success Scenario:

1. City Operator opens up SCEMAS on their personal device and logs in.
2. System authenticates user.
3. Operator accesses operational dashboard and clicks on anomalous trend violations.
4. System highlights active anomalous trend violations in the alerts section.
5. Operator selects specific violation to view in more detail.
6. System displays violation details.
7. Operator acts on violation.
8. System updates the alert status to “acknowledged”.

Secondary Scenario:

2i. System fails to authenticate user.
4i. No anomalous trend violations are active.
7i. Operator does not act on violation.
8i. System fails to update alert status.

VP3. Government (Admin)

Main Success Scenario:

1. Government Official opens up SCEMAS and logs in as an official.
2. System authenticates user.
3. Official accesses operational dashboard and clicks on anomalous trend violations.
4. System highlights active anomalous trend violations.
5. Official acts on violation.
6. System updates alert status to “acknowledged”.

Secondary Scenario:

2i. System fails to authenticate user.
4i. No anomalous trend violations are active.
    4i.1. Official does not perform any actions.

VP4: Sensor Company

N/A

VP5: Data Analysts

Main Success Scenario

1. Data Analyst opens SCEMAS and enters log in details.
2. System authenticates analyst.
3. Analyst accesses data the system has collected (historical and recent).
4. System displays numerical data.
5. Analyst creates an algorithm to check if trend is anomalous.
6. Analyst concludes that trend is not anomalous.

Secondary Scenario:

2i. System fails to authenticate user.
5i. Analyst fails to create algorithm.
6i. Analyst concludes trend is anomalous.
    6i.1. Analyst sends an alert through the system.

VP6: Developers

N/A

Global Scenario

Pre-Condition: An anomalous threshold has been defined and enabled.

Main Success Scenario:

1. User opens up SCEMAS on their personal device and selects type of user.
2. System authenticates user.
3. User accesses operational dashboard and clicks on anomalous trend violations.
4. System highlights active anomalous trend violations in the alerts section.
5. User selects specific violation to view in more detail.
6. System displays violation details.
7. User acts on violation.

Secondary Scenario:

2i. System fails to authenticate user.
3i. User instead accesses data collected from sensors.

4i. data shows that trend is anomalous,

4i.1 user sends an alert through the system

5i. No anomalous trend violations are active.

7i. User does not act on violation.

7i.1 system keeps violation marked as unresolved

BE8. Validating Data

Precondition: Sensors have been set up and installed by the sensor company.

VP1: General Public

N/A

VP2: City Operators

N/A

VP3: Government (Admin)

Main Success Scenario:

1. Government official opens up SCEMAS on their device and logs in.
2. Official accesses the operational dashboard.
3. System displays options on the dashboard.
4. Official navigates to trends and analytics section.
5. System displays environmental data produced by sensors.
6. Official analyzes data and confirms it follows MQTT protocol.

Secondary Scenario:

6i. Official confirms that data does not follow MQTT protocol.

6i.1. Official sends an alert to Data Analysts through the dashboard.

6i.2. System stores and delivers the message.

6i.3. System logs the message event.

VP4: Sensor Company

Main Success Scenario:

1. Message from analyst is sent to the sensor company using external methods.

Secondary Scenario:

N/A

VP5: Data Analysts

Main Success Scenario:

1. Analyst opens up SCEMAS on their device and logs in.
2. Analyst navigates to operational dashboard.
3. System displays options on the dashboard.
4. Analyst clicks on the trends and analytics section.
5. System displays historical and real time environmental trends.
6. Analyst reviews charts and data and confirms that values are within expected range.

7. Analyst confirms sensor data seems normal.

Secondary Scenario:

1i. Analyst views alert from government official about the data not following MQTT protocol
    1i.1. Analyst reviews the flagged data and replies to the Government Admin through the Messaging Service
    1i.2. If sensor investigation is needed, the Data Analyst contacts sensor company externally.
6i. Analyst concludes that values are not within expected range.
    6i.1. Analyst messages a member from the sensor company through external contact methods.

VP6: External Developers

Main Success Scenario:

1. Developer opens up SCEMAS and logs in
2. System displays options on the operational dashboard
3. Developer accesses public API requests
4. System displays list of recent API requests
5. Developer selects one and checks if the data requested has been validated
6. Developer confirms data has been validated and accepts the public API request

Secondary Scenario:

4i. No APIs are currently requested
6i. Data has not been validated
    6i.1 Developer rejects the API request

Global Scenario:

Precondition: Sensors have been set up and installed by the sensor company

Main Success Scenario:

1. User opens up SCEMAS on their device and logs into system.
2. User accesses operational dashboard.
3. System displays options on dashboard based on the user.
4. User clicks on trends and analytics section.
5. System displays historical and real time environmental trends.
6. User reviews charts and data and confirms that values are within expected range.
7. System indicates data status is validated.
8. User proceeds with their role specific action.

Secondary Scenario:

6i. User concludes that values are not within expected range.
    6i.1. User contacts an authorized individual for repair.
    6i.2. Authorized individual accesses system to see messages.
6ii. Data does not follow MQTT protocol.

6ii.1. Alert is sent out in the system.
6iii. Data has not been validated.
7i. System keeps data unvalidated.

5 Non-Functional Requirements

5.1 Look and Feel Requirements

5.1.1 Appearance Requirements

LF-A1. The system will be designed to use generalized non-offensive colouring schemes.
Rationale: The colour scheme should be widely appealing, without leaning into any background to maximize compatibility and flexibility with users [3].

LF-A2. The system will avoid using any excessively bright or saturated colours.
Rationale: Serves to minimize eye strain, and to create a more visually appealing application [4].

LF-A3. The system will avoid needlessly complex designs.
Rationale: By steering towards minimal complexity, the software will massively increase its usability and flexibility within a large range of technological knowledge [3].

LF-A4. Interactable elements should have consistent functionality and appearance.
Rationale: Greatly increases user experience, while reducing overall confusion on all levels of technology nuance [3].

5.1.2 Style Requirements

LF-S1. The system must clearly indicate when certain events, alarms, or other miscellaneous events trigger.
Rationale: This is essential for a responsive user experience, greatly improving the feel of the software [5].

LF-S2. The system shall never go over 5 interactable elements at one given state.
Rationale: Having a surplus of interactable elements may lead to user confusion, by limiting the possible actions at any given point will reduce it [3].

LF-S3. The system must scale correctly to a window, or to variable screen sizes.
Rationale: When software does not correctly scale to the environment around it, it ends up feeling cheap and unsatisfying to use [3].

LF-S4. The system should have a uniform visual design overall.
Rationale: This is to avoid confusion, and to give a feeling of cohesion throughout the software [3].

5.2 Usability and Humanity Requirements

5.2.1 Ease of Use Requirements

UH-EOU1. The operational dashboard should allow City Operators to view active alerts and open the alert page in 3 clicks/keypresses or fewer from the main dashboard page.
Rationale: Operators need to reach critical information quickly during

incidents, and minimizing steps reduces delay and mistakes [6].

UH-EOU2. The public dashboard should display the current overall status of a selected zone using labels like “Good”, “Moderate”, and “Poor”, as well as a short explanation on what the label means [7].
Rationale: Public users may not be able to understand the raw sensor values, so the system should be able to summarize findings clearly to improve understanding and trust.

UH-EOU3. Both dashboards should provide a zone search or filter feature that allows users to find a zone by name or status within 10 seconds under normal conditions.
Rationale: The dashboard must provide city data across various geographical zones using maps, so users will need an efficient way to locate zones quickly [1]. Providing a search feature gives a clean interface for users.

UH-EOU4. When a user acknowledges an alert, the system should display a confirmation message while updating the alert status within 7 seconds.
Rationale: Users need to know right away if their alert acknowledgement worked. A fast status update keeps the dashboard feeling responsive during critical events [8].

5.2.2 Personalization and Internationalization Requirements

UH-PI1. The operator dashboard should allow users to select their default view like “Alerts”, “Map”, or “Trends” that stays even when the user logs out.
Rationale: Different operators use the dashboard for different purposes, so having a preferred view reduces repeated setup and makes the dashboard faster to use [9].

UH-PI2. The public and operational dashboard should allow users to save up to 4 zones for quick access.
Rationale: People usually check the same areas repeatedly, like their home, school, or workplace, so this makes the system faster to use [9].

UH-PI3. The system should have the dates and times of the city being monitored as well as the difference from the user’s time zone.
Rationale: Time zones can be confusing especially when it greatly differs from the user's zone [10].

5.2.3 Learning Requirements

UH-L1. The system should have a short “Help” page that explains key terms used in SCEMAS like zone, threshold, and severity.
Rationale: A designated page that keeps vocabulary meaning clear reduces confusion and helps new users learn the platform faster [11].

UH-L2. The public dashboard should have short explanations of how the status of areas are determined in plain language.
Rationale: People will trust a dashboard if they understand how labels are determined and what they mean [12].

5.2.4 Understandability and Politeness Requirements

UH-UP1. Error messages shown to users explain what went wrong and what the user can do next rather than technical error terms like HTTP codes.
Rationale: Users will not be able to solve problems if the message is confusing or too technical [13].

UH-UP2. The system should not use language that blames the user but rather neutral wording like “We could not verify your login”.
Rationale: Polite wording improves user trust and reduces frustration [13].

UH-UP3. When telemetry data is missing or rejected, the operational dashboard should display that data is unavailable and provide a high-level reason, without exposing internal details.
Rationale: Operators need to understand why data is unavailable so they can respond appropriately [13].

5.2.5 Accessibility Requirements

UH-A1. The dashboards should be able to be navigated only using the keyboard for main tasks like logging in, selecting zones, and viewing alerts.
Rationale: Some users rely on keyboard navigation, and it remains useable for everyone [14].

UH-A2. The dashboards shall not rely on colour only to communicate important status. They should use text and icons as well for alert severity and zone status.
Rationale: Alerts with only colours fails for colourblind users and low-quality displays [15].

UH-A3. The system shall allow users to increase text size up to 200% without losing access to core features.
Rationale: Users will poor vision need larger text while still being able to use the system [15].

5.3 Performance Requirements

5.3.1 Speed and Latency Requirements

PR-SL1. The system shall process incoming telemetry data and update internal zone-level aggregates (e.g., 5-minute averages, hourly maximums, etc.) within 10 seconds of receiving valid sensor data [1].
Rationale: The purpose of SCEMAS is to function as a real-time monitoring system. To facilitate prompt decision-making by city operators, timely aggregation guarantees that dashboards, alarms, and public summaries represent actual environmental conditions.

PR-SL2. The operational dashboard shall display a new critical alert to City Operators within 30 seconds of the alert being detected by the system [1].
Rationale: This requirement directly supports the system's goal of allowing operators to view and recognize critical alerts within 30 seconds, allowing for a quick response to potentially hazardous environmental conditions.

5.3.2 Safety-Critical Requirements

PR-SC1. The system shall prioritize critical alert generation and delivery over non-critical background tasks when a critical environmental threshold is violated or an abnormal trend is recognized.
Rationale: Environmental alerts may be related to public health and safety. Prioritizing alert processing decreases the likelihood of delayed reactions in dangerous conditions.

5.3.3 Precision or Accuracy Requirements

PR-PA1. The zone-level environmental metrics displayed on dashboards and exposed via a public API must be computed using validated telemetry data and must accurately reflect the defined aggregation window (e.g., 5-minute average, hourly maximum, etc.) [1].
Rationale: Accurate aggregation is critical for maintaining system confidence and ensuring that operators, analysts, and the public make informed decisions based on reliable data.

PR-PA2. Alert timestamps stored and displayed by the system must be accurate to within one second of the time the threshold violation is detected [16].
Rationale: Precise timestamps are required for auditing, post-incident investigation, and comparing reports to previous environmental trends.

5.3.4 Reliability and Availability Requirements

PR-RA1. The backend services must have an uptime of 99.95% [17].
Rationale: SCEMAS relies heavily on continuous monitoring. A high level of availability guarantees that environmental data collection and alerting features remain operational most of the time.

PR-RA2. The system shall store all valid telemetry and alert data in secure and durable data storage to ensure that no alert is lost due to system failure [18].
Rationale: Loss of alert data may hinder operators from responding effectively, jeopardizing the system's reliability and auditability.

5.3.5 Robustness or Fault-Tolerance Requirements

PR-RFT1. The system shall reject all invalid telemetry data without affecting the processing of the valid telemetry data [1].
Rationale: Sensors may fail or transmit corrupted data. The system must be stable and continue to handle valid data without interruption.

PR-RFT2. The system must continue ingesting data, creating alerts, and perform aggregation operations, even if dashboard or public API service is temporarily unavailable [1].
Rationale: Temporary interface failures should not affect core monitoring and alerting functionality.

5.3.6 Capacity Requirements

PR-C1. The system shall support simultaneous ingestion and processing of data from at least 100 sensors without alert detection performance degradation [19].

Rationale: This capacity represents a realistic small-city deployment for the project's scope and enables the system architecture to be tested with concurrent data streams.

5.3.7 Scalability or Extensibility Requirements

PR-SE1. The system architecture must allow the addition of new environmental metrics, sensors, or geographic zones, without requiring major changes to the system architecture [20].
Response: Smart city systems evolve over time. Designing SCEMAS for scalability ensures that it can accommodate future growth while minimizing rework.

5.3.8 Longevity Requirements

PR-L1. The system shall retain historical telemetry and alert data for a minimum of 90 days for analysis, auditing, and trend review [2].
Response: Access to historical data enables long-term environmental study, post-event inquiries, and system evaluation without necessitating permanent storage.

5.4 Operational and Environmental Requirements

5.4.1 Expected Physical Environment

OE-EPE1. The system will support integration with real IoT sensors that securely and accurately transmit environmental data [21].
Rationale: System depends on external sensors infrastructure for accurate reading and data upon which it operates.

5.4.2 Requirements for Interfacing with Adjacent Systems

OE-IA1. The system shall interact with MQTT based IoT sensors [21].
Rationale: System needs sensors to acquire data from.

OE-IA2. System will support integration with external SMS or notification service for alert notification.
Rationale: Environmental amber alert type system requires integration with a messaging gateway to deliver alerts in real time [21].

5.4.3 Productization Requirements

N/A

5.4.4 Release Requirements

OE-R1. SCEMAS shall ensure that updates to dashboards or alerting logic do not disrupt core telemetry ingestion and alert processing services [21].
Rationale: Environmental monitoring must remain operational even during system updates.

5.5 Maintainability and Support Requirements

5.5.1 Maintenance Requirements

MS-M1. System will allow threshold modifications through the administrative dashboard without necessary system redeployment [22].
Rationale: Policy changes may elicit threshold adjustments, and these must be managed without system requiring downtime

5.5.2 Supportability Requirements

MS-S1. System shall maintain audit logs of authentication events, changes in threshold and alert acknowledgement for a minimum of 120 days [23].
Rationale: Logs assist in troubleshooting, compliance verification and system transparency

5.5.3 Adaptability Requirements

MS-A1. System will support addition of new environmental metrics without system needing refactoring [22].
Rationale: Monitoring types and wanted information change over time and system must accommodate these changes

5.6 Security Requirements

5.6.1 Access Requirements

SR-AC1. The app must allow the user to access their account if and only if the log in details given by them are correct [24].
Rationale: This is to ensure that all the accounts accessing the app are registered.

SR-AC2. The system must support role-based access control for different user types [25].
Rationale: Different users require different levels of access. For example, general users may only view data, while government operators and analysts may upload, modify, or analyze datasets.

5.6.2 Integrity Requirements

SR-INT1. The user data stores within the app should be encrypted [26].
Rationale: To ensure user data is not stolen, data encryption is necessary.

SR-INT2. All data must be validated before being able to be viewed by the public [27].
Rationale: Input validation prevents corrupted or malicious data from compromising system integrity.

5.6.3 Privacy Requirements

SR-P1. The app must inform and ask users for the storage of their information [28].
Rationale: This is part of the Google Play Developer Distribution agreement.

SR-P2. The app must provide users a legally adequate privacy notice [28].
Rationale: This is part of the Google Play Developer Distribution agreement.

5.6.4 Audit Requirements

SR-AU1. The app must log the user's app usage data [29].
Rationale: This is to gather information on how the users use the app and to fix any potential bugs that users might face during their usage

SR-AU2. The system must log authentication events such as logins, logouts, and failed login attempts [29].
Rationale: These logs help detect suspicious activity and support security audits.

5.6.5 Immunity Requirements

SR-IM1. The system must not accept unexpected input [30].
Rationale: This is to prevent attacks like SQL Injections

5.7 Cultural and Political Requirements

5.7.1 Cultural Requirements

CP-C1. The public dashboard will aim to avoid slang, idioms, or region-specific phrases and expressions.
Rationale: SCEMAS is aimed to work in a variety of environments, across many different cultural backgrounds [31].

CP-C2. The software should avoid phrases, imagery, or innuendos that could be considered offensive where SCEMAS is available, through utilization of a list of appropriate icons and images to refer to.
Rationale: SCEMAS should avoid creating public controversy and offending users through any designs within the system [32].

CP-C3. SCEMAS will use inclusive, non-accusatory language when sending messages or alerts to users.
Rationale: Environmental conditions can disproportionately affect communities, so neutral wording can help to prevent social stigmas on select groups and improve community relations [32].

5.7.2 Political Requirements

CP-P1. The public dashboard shall not provide endorsement or criticism of any political party, candidate, or policy.
Rationale: SCEMAS should remain as a politically neutral software, as politics is not the aim of the average use case [32].

CP-P2. SCEMAS will enforce role-based separation of duties for actions that are considered politically sensitive or have high-impact.
Rationale: Role separation reduces risks through modularizing access levels [33].

5.8 Legal Requirements

5.8.1 Compliance Requirements

LR-COMP1. SCEMAS should not collect, process, or store personally identifiable information (PII) like names, emails and phone numbers.
Rationale: The system is used for environmental monitoring and by avoiding PII, it reduces privacy and legal risks [1].

LR-COMP2. The public dashboard and public API should only display zone-level summary data that is approved for the public and not provide raw sensor readings.
Rationale: Showing summarized data protects privacy and decreases the possibility of someone figuring out sensitive details about the system [1].

LR-COMP3. SCEMAS should display a small privacy message that states what data is being collected, not collected, and how it is being used.
Rationale: Users must be clearly informed to build trust and meet privacy expectations [34].

LR-COMP4. SCEMAS should restrict access to operator features using role-based access control (RBAC) and require authentication before accessing non-public data like raw telemetry, device health, and alert logs.
Rational: This helps keep sensitive data private and prevent unauthorized access [1].

LR-COMP5. SCEMAS should keep its security logs like login attempts, role changes, and authentication attempts for at least 120 days.
Rationale: Audit logs make it easier to see who did what and allows the ability to review what happened if something goes wrong [35].

5.8.2 Standards Requirements

LR-STD1. SCEMAS should use encryption during transit for all data that contains telemetry, login details, or system data by using TLS, including MQTT over TLS.
Rationale: This ensures that others in the network cannot read or modify data while it is being sent [1].

LR-STD2. LR-STD2. The public API should follow REST conventions by providing read-only endpoints and return structured responses and errors.
Rationale: By following REST API conventions, it makes it easier for third parties to use the API correctly [1].

LR-STD3. SCEMAS should have rate limiting for the public API and return a message when the limit is exceeded.
Rationale: This helps overuse, data-scarping, and abuse of the public API [1].

LR-STD4. SCEMAS should not store any user passwords in plain text and should secure credentials using security practices like single-sign-on (SSO).
Rationale: Protecting user credentials helps reduce the risk of data breaches and account compromise [36].

6 Innovative Features

The chosen innovative feature to implement is an Amber Alert style environmental notification system. This feature will send direct text notifications to users when a severe environmental threshold has been exceeded. It would be convenient for users who are not actively monitoring the dashboard but still need immediate awareness of dangerous conditions. This may increase user trust and engagement with the system while improving overall public safety response.

We have also considered additional innovative features as listed below:

- Air quality recommendations (even if thresholds are not exceeded but public health guidelines suggest precaution)
- Visualization of the operating environment (map view, globe view, interactive overlays)
- Recommendations for government or city operators on which services to dispatch
- Environmental data visualizer (e.g., showing how poor air or water quality visually appears)
- Mobile application version of SCEMAS
- Pinned locations for constant monitoring
- Ranked environmental quality comparisons (e.g., between city zones)
- Severity-based notification tiers (e.g., moderate vs critical alerts)
- Optional subscription model for advanced alert features

A Division of Labour

Include a Division of Labour sheet which indicates the contributions of each team member. This sheet must be signed by all team members.

Aakash Satishkumar:

- 1.2 Scope
- 2.2 Product Functions
- Section 4 BE1 and BE4
- 5.2 and 5.8

Ali Shareeff:

- 2.3 User Characteristics
- 2.4 Constraints
- 4 BE7, BE8
- 5.6

Harmanpreet Singh Sagar

- 1.5 Overview
- 2.6 Apportioning of Requirements
- 3 Use Case Diagram
- 4 BE3
- 5.3 Performance Requirements
- Formatting document

Jason Kim

- 1.1 Purpose
- 2.1 Product Perspective
- 4 BE5, BE6

- 5.1, 5.7

Praneet Singh

- 2.5 Assumptions and Dependencies
- 4 BE2
- 5.4
- 5.5
- 6 Innovative Feature